OpenCPU SDK

SSL Configuration Notes

Issue 1.0
Date 2025-01-23

Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

THIS DOCUMENT PROVIDES SUPPORT FOR PRODUCT DESIGN BY THE USER. THE USER MUST DESIGN AND DEBUG THE PRODUCT ACCORDING TO THE SPECIFICATIONS AND PARAMETERS IN THIS DOCUMENT. NEOWAY ASSUMES NO RESPONSIBILITY FOR PERSONAL INJURY AND PROPERTY LOSS CAUSED BY USER'S IMPROPER OPERATION.

DUE TO PRODUCT VERSION UPDATES OR OTHER REASONS, THE CONTENT OF THIS DOCUMENT WILL BE UPDATED AS NECESSARY WITHOUT PRIOR NOTICE.

UNLESS OTHERWISE AGREED, ALL STATEMENTS, INFORMATION AND SUGGESTIONS IN THIS DOCUMENT CONSTITUTE NO WARRANTY WHETHER EXPRESS OR IMPLIED.

Neoway Technology Co., Ltd. provides customers with comprehensive technical support. For any inquiry, please contact your account manager directly or send an email to the following email address:

Sales@neoway.com

Support@neoway.com

Website: http://www.neoway.com

Contents

About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for system engineers (SEs), development engineers, and test engineers.

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Gao Zheng

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document describes the SSL configuration file, nwy_ssl_config.h.

2 Data Structure

2.1 nwy_ssl_version_e

```
typedef enum{
   NWY_VERSION_SSL_V3_E,        //SSL3.0
   NWY_VERSION_TLS_V1_0_E,      //SSL1.0
   NWY_VERSION_TLS_V1_1_E,      //SSL1.1
   NWY_VERSION_TLS_V1_2_E,      //SSL1.2
   NWY_VERSION_MAX_E
}nwy_ssl_version_e;
```

2.2 nwy_ssl_auth_mode_e

```
typedef enum{
   NWY_SSL_AUTH_NONE_E = 0,   //No authentication
   NWY_SSL_AUTH_ONE_WAY_E,  /*Manage server authentication
   NWY_SSL_AUTH_MUTUAL_E,   //Manage server and client authentication if requested by the
remote server*/
   NWY_SSL_AUTH_MAX_E
}nwy_ssl_auth_mode_e;
```

2.3 nwy_ssl_cert_t

```
typedef struct
{
  char *cert_data;
int cert_len;
}nwy_ssl_cert_t;
```

2.4 nwy_ssl_conf_t

```
typedef struct
{
  nwy_ssl_version_type_e ssl_version;
  nwy_ssl_auth_mode_e authmode;
  nwy_ssl_cert_t cacert;
  nwy_ssl_cert_t clientcert;
  nwy_ssl_cert_t clientkey;
  int sni_name_size;    /**< Length of the SNI server name. */
  char *sni_name; /**< Server name for SNI. */
```



```
}nwy_ssl_conf_t;
```


3 Solving Certificate Issues for Multi-Domain Servers

3.1 SNI Extension

SNI (Server Name Indication) is an extension to enhance SSL (Secure Socket Layer) and TLS (Transport Layer Security) interactions between servers and clients. It addresses the limitation where a server could previously only use one certificate for one domain. With virtual hosting support, a single server can now serve multiple domains, making SNI essential to meet this requirement.

3.2 SSL Certificate Validation Process

During SSL certificate validation, a check is performed: the browser compares the entered address with the certificate's name. If they match, validation proceeds; if not, the certificate is deemed untrustworthy.

For example, consider a company with two domains: www.123.com and www.567.com, both pointing to the same IP address, 222.12.34.56. If the server’s certificate is for www.123.com, users accessing it via www.567.com will encounter a trust issue with the server's certificate. This discrepancy causes clients to distrust the server for one of the domains.

3.3 Solution

To resolve this, the client includes the server name extension in the Client Hello message (note: this extension is absent when accessing via an IP address). The client provides the domain name in the server name field. The server then parses the server name and selects the appropriate certificate corresponding to the specified domain name.

